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REVIEW ARTICLE



Efficacy of fresh frozen plasma transfusion in comparison with conventional regimen in organophosphate poisoning treatment: a meta-analysis study

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ABSTRACT

Objective: To evaluating the efficacy of fresh frozen plasma (FFP) in comparison with conventional regimen in the treatment of organophosphate (OP) poisoning.

Methods: PubMed, ScopeMed, Cochrane, Scopus, and Google Scholar databases were searched. The search strategy used the following key words “organophosphate” and “poisoning or toxicity”, “(atropine and oxime)”, “fresh frozen plasma”, “clinical trial”, “outcome”. The treatment with atropine or/and oxime was considered conventional therapy. The length of hospitalization, the length of ICU admission, need for mechanical ventilation and its duration, clinical recovery point, choline esterase level, mortality rate, and intermediate syndrome (IMS) occurrence were the key outcomes of interest. Databases were searched during the period of 2003–2019. Five studies were included in the analysis.

Results: Pooling of data showed that the relative risk (RR) of mortality in OP poisoning for five included trials comparing FFP-treated group with conventional regimen therapy was [0.563 (95% CI (0.252, 1.255))]. The summary of RR for IMS in two studies was [RR: 1.34, 95% CI (0.655, 2.742)]. In addition, there was a non-significant mean difference (MD) in hospital stay [MD: −0.106, 95% CI (−0.434, 0.223)] in three included trials. A significant MD was observed in the length of ICU admission in two trials between FFP-treated group compared to the conventional treatment group [MD: −2.672, 95% CI (−4.189, −1.154)], but after random effects meta-analysis, the changes were not significant [MD: −2.015, 95% CI (−6.308, 2.277)]. The summary of fixed-effect meta-analysis for choline esterase level in three trials was [MD: −0.117, 95% CI (−0.468, 0.234)]. The RR of ventilation requirement for two included trials in the FFP-treated group comparing to the conventional regimen therapy was [0.84, 95% CI (0.691, 1.022)] while for ventilation duration in two studies was [MD: −0.183, 95% CI (−0.567, 0.201)].

Conclusion: The addition of FFP to conventional therapy did not improve the outcomes of mortality, IMS, hospital length of stay, cholinesterase levels, need or duration of mechanical ventilation, and only the length of ICU stay could affect in the treated group.

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Introduction

Organophosphorus pesticides are used broadly for farming and domestic purposes. Self-poisoning by organophosphorus compounds is a crucial clinical problem in rural parts of developing countries that kills about 200,000 people each year (Güven, Sungur, Eser, et al. 2004; Eddleston et al. 2008). Pesticide toxicity is the main cause of approximately 60% of the estimated 500,000 deaths from self-harm in the suburbs each year (World Health Organization 2002; Eddleston and Phillips 2004). Although accidental poisoning is responsible for a fewer number of deaths, there is more mortality in areas where organophosphorus pesticides exist (World Health Organization 1990). It is suggested that immediate resuscitation with atropine, oxygen, respiratory support, and fluids is required to enhance oxygen delivery to tissues (Eddleston et al. 2008).

It is not clear how oximes work, whether they are only useful for patients who are poisoned by specific pesticides or those with moderate poisoning (Eyer 2003). A number of studies have shown that atropine and oximes cannot prevent morbidity and mortality in poisoned patients (Masson et al. 1997; Johnson et al. 2000; Sudakin et al. 2000; Eddleston et al. 2002; Sivagnanam 2002). Generally, oximes are used widely in treating organophosphorus poisoned humans because extensive studies on oximes have shown its protective effects against organophosphorus *in vitro*. However, the results of *in vivo* studies on humans are not proved yet and there are no exhaustive meta-analyses to indicate oximes effectiveness (Pawar et al. 2006; Peter et al. 2006; Pazooki et al. 2011). It is only demonstrated that oximes may be useful in moderate to severe poisonings in which the patient has been referred to hospital and received treatment promptly (less than 6 h of exposure) (Pawar et al. 2006). In addition, other studies have shown that oximes are ineffective in organophosphate (OP) poisoning management and also may cause harm. Accordingly, clinicians should be cautious about using oximes in poisoned patients (Eddleston et al. 2002; Peter et al. 2006), and other treatment regimens for organophosphorus poisoning are needed (Peter et al. 2008).

Different strategies have been trialed for organophosphorus poisoning treatment, namely magnesium, hemoperfusion, alkalinization, N-acetyl cysteine, diazepam, clonidine, and adenosine receptor antagonists (Peter et al. 2008). Since the goal of treatment is negating the toxins, also using

bioscavengers as a treatment has played an important role in ameliorating the damage caused by OPs (Güven, Sungur, et al. 2004; Güven, Sungur, Eser, et al. 2004; Pichamuthu et al. 2010). Bioscavengers therapy includes utilizing enzymes such as cholinesterase which is capable of blocking OP compounds before reaching the toxic threshold (Doctor and Saxena 2005). Fresh frozen plasma (FFP) has been investigated in human studies as a source of butyrylcholinesterase (BuChE) to treat organophosphorus poisoning (Güven, Sungur, et al. 2004; Güven, Sungur, Eser, et al. 2004; Pichamuthu et al. 2010).

For the first time, the effect of plasmapheresis using FFP on an OP-poisoned patient was examined by Güven, Sungur, Eser, et al. (2004) on 12 patients in a partially randomized controlled study, and the results were compared with 21 patients receiving conventional treatment. However, the results of another study in 2010 did not approve the efficacy of FFP in patients with OP poisoning (Pichamuthu et al. 2010). Since using FFP is a method of treatment in OP poisoning which has revealed opposing results in the previous few studies, we compared the effect of FFP therapy with conventional regimen in OP-poisoned patients in a systematic review with meta-analysis.

Materials and methods

Data sources

A systematic review was conducted in electronic sources including ISI, PubMed, ScopeMed, Cochrane, Scopus, and Google Scholar databases for the studies that investigated the efficiency of FFP in the treatment of OP poisoning. This was performed over the time period of 2003–2019. The search strategy used the following key words “organophosphate” and “poisoning or toxicity”, “(atropine and oxime)”, “fresh frozen plasma”, “clinical trial”, “outcome”. These were limited to the clinical studies in humans. Each included article had a secondary search of their respective references for any additional studies. The treatment with atropine or/and oximes was considered conventional therapy. Studies in which the FFP and conventional therapy were compared with the parallel control group who only received the conventional therapy were taken into consideration. Data were extracted regarding the types of therapeutic regimens. The main outcome of interests was the length of hospitalization, ICU admission incidence, mechanical ventilation utilization and its duration, clinical recovery point, choline esterase level (IU/L), mortality rate, and intermediate syndrome (IMS) occurrence.

Eligibility criteria for considering studies for this review

Four reviewers independently examined the title and abstract of each article to eliminate duplicates, reviews, case studies, and studies without control group. Disagreements were resolved by consensus. Studies with English and Persian language were included and if they did not provide data that allowed the investigators to calculate standard deviation (SD)/standard error (SE) for effect estimates, and then they

Table 1. Jaded quality score of trial included in the meta-analysis.

Studies	Randomization	Blinding	Withdrawal described	Total
Saad et al. (2017)	2	−1	1	2
Güven, Sungur, et al. (2004)	1	−1	0	0
Pichamuthu et al. (2010)	2	−1	1	2
Pazooki et al. (2011)	2	−1	1	2
Dayananda et al. (2016)	2	−1	1	2

were excluded. Study participants with OP poisoning who received FFP plus conventional therapy vs. sole conventional therapy were included. Only manuscripts with full-text accessibility were included in the final analysis.

Assessment of trial quality

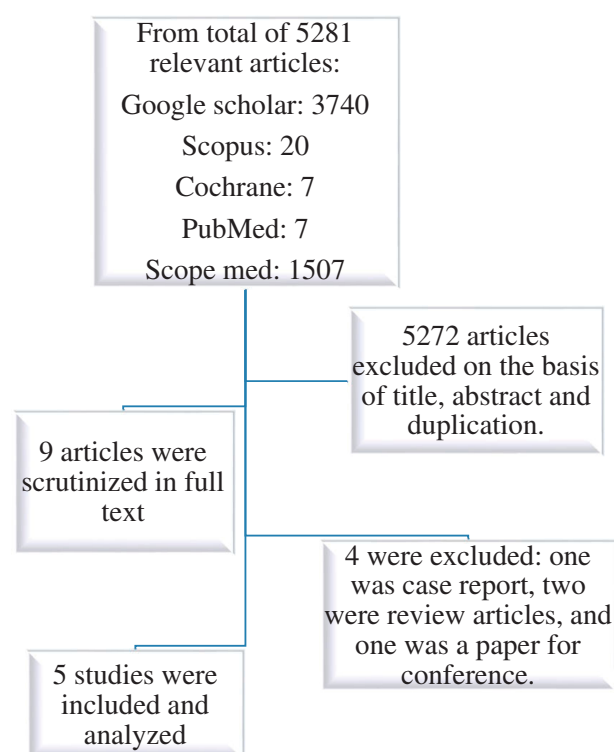
The quality of studies was determined based on their description of randomization, blinding, and dropouts (Jadad et al. 1996; Table 1). This is summarized as follows: (a) whether participant was randomized or not (yes = 1 point, no = 0), (b) whether randomization was described appropriately or not (yes = 1 point, no = 0 point), (c) double-blinding occurred (yes = 1 point, no = 0), (d) double-blinding was described appropriately (yes = 1 point, no = 0), (e), whether withdrawals and dropouts were described or not (yes = 1 point, no = 0). The quality scale ranges from 0 to 5 points, with a low-quality report receiving a score 2 or less and a high-quality report receiving a score of at least 3.

Data collection

Data were extracted independently by four reviewers using a standard form designed by the researchers and then cross-checked. In the case of missing information in the included studies, investigators sent an email to the responsible author of the article

Statistical analysis

Overall weighted mean difference (WMD) and 95% confidence interval (CI) were calculated for the continuous outcomes. If the same continuous outcome is measured differently across studies, an overall standardized mean difference (SMD) and 95% CI were calculated. Otherwise, for the categorical variables, the relative risk (RR) and 95% CI were calculated using the Der Simonian–Laird (for random effects) method. Standardized effect size (ES) and 95% CI were calculated using the Der Simonian–Laird (for random effects) method. Data were meta-analyzed if possible according to the difference of recorded variables. Meta-regression or subgroup analysis was used in the case of statistical heterogeneity. The Cochrane Q test was used to test heterogeneity and $p < 0.05$ considered significant. In the case of heterogeneity or few included studies, the random effects model was used. Statistical analyses were carried out with comprehensive Meta-analysis Software, version 2.0 (BioStat Inc., Englewood, NJ, USA).

**Figure 1.** Flow diagram for study selection.

Assessment of publication bias

Potential publication bias was evaluated through visual examination of the funnel plot and by the Egger's regression asymmetry test. The funnel plot graphically represents the distribution of the retrieved studies considering ES and SE. It is used for visually detecting potential publication bias based on the symmetry of the plot.

Assessment of risk of bias in included studies

Methodological quality was assessed independently by at least four reviewers according to the Cochrane Collaboration Handbook and Cochrane Review guidelines. Reviewers were independently assessed the risk of bias within each included study based on the following domains with ratings of "Yes" (low risk of bias); "No" (high risk of bias) and "Unclear" (uncertain risk of bias).

Sensitivity analysis

Sensitivity analyses were conducted to assess the impact of the study quality. A sensitivity analysis compares studies fulfilling and not fulfilling the "Quality" criteria and asks the question, "Are the findings robust to the decisions made in the process of obtaining them?" This involves the removal of studies that meet certain criteria (e.g. poor quality, commercial sponsorship, conference abstract) to determine their effect on the overall result. These were undertaken by including: (1) only those with low risk of selection bias (associated with sequence generation or allocation concealment); (2) Only those with low risk of performance bias (associated with

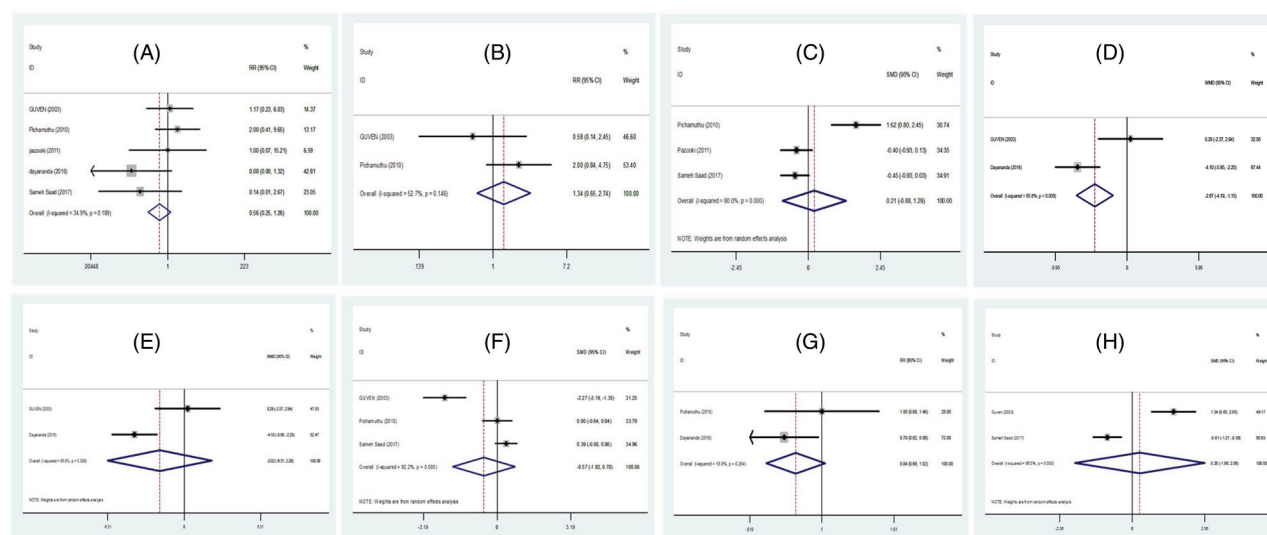


Figure 2. (A) Random effects meta-analysis of the relative risk in mortality rate comparing FFP treatment with conventional therapy. (B) Fixed effects meta-analysis of the relative risk in intermediate syndrome comparing FFP therapy with conventional regimen. (C) Random effects meta-analysis of the mean difference in length of hospital stay comparing FFP therapy with conventional regimen. (D) Fixed effects meta-analysis of the mean difference in length of intensive care unit stay comparing fresh frozen plasma therapy with conventional regimen. (E) Random effects meta-analysis of the mean difference in length of intensive care unit stay comparing fresh frozen plasma therapy with conventional regimen. (F) Random effects meta-analysis of the mean difference in serum choline esterase level comparing fresh frozen plasma therapy with conventional regimen. (G) Fixed effects meta-analysis of the mean difference in need for mechanical ventilation, comparing FFP therapy with conventional regimen. (H) Random effects meta-analysis of the mean difference in ventilation duration comparing FFP therapy with conventional regimen.

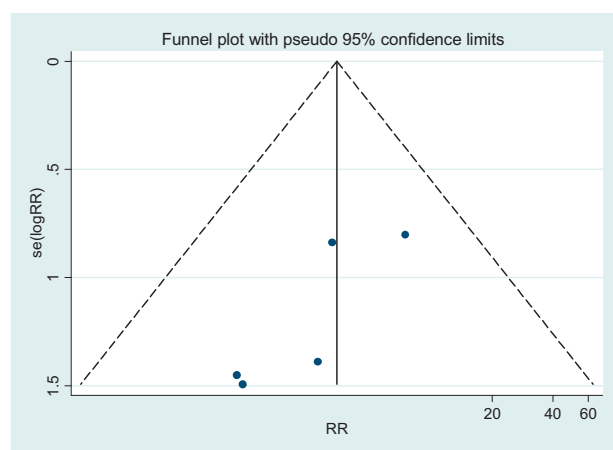


Figure 3. Funnel plot for random effects meta-analysis of RR in mortality rate comparing FFP treatment with conventional therapy.

issues of blinding); (3) Only those with low risk of attrition bias (associated with completeness of data).

Results

We reviewed a total of 5281 abstracts and titles; 5272 were excluded based on duplication or title and abstract irrelevancy. Therefore, nine articles were scrutinized in full text, of which, five articles met inclusion criteria and were considered eligible for systematic review. One of the excluded studies was a case report (Vučinić et al. 2013), two were review articles (Peter et al. 2007; Balali-Mood and Saber 2012) and the one was a conference paper (Nikfar and Abdollahi 2012; Figure 1). The quality of clinical trials was assessed using the Jadad score. Four studies received 2 while the other one was scored 0 (Table 1). Outcomes according to included studies

have been provided in Table 2. Characteristic of included studies containing study type, number of patients, intervention and outcome assessing are presented in Table 3.

Effect of FFP therapy in comparison to conventional regimen on mortality in patients with OP poisoning

The summary for RR for mortality in OP-poisoned patients for five selected trials (Güven, Sungur, Eser, et al. 2004; Pichamuthu et al. 2010; Pazooki et al. 2011; Dayananda et al. 2016; Saad et al. 2017) comparing FFP to conventional treatment was 0.563 with a [95% CI (0.252–1.255), $I^2 = 34.9\%$ ($p = 0.189$)] (Figure 2(A)). As seen in Figure 2(A), there was a minor decrease in FFP groups in mortality rate compared to the conventional treated groups, but it was not statistically significant. Thus, the inconsistent results and publication bias (Figure 3) in terms of mortality between FFP therapy in comparison to conventional therapy was observed, although based on Begg's ($p = 0.050$) funnel plot and Egger test ($p = 0.074$), we did not detect any noteworthy difference.

Effect of FFP therapy in comparison to conventional regimen on IMS in patients with OP poisoning

The summary for RR of IMS in OP poisoning patients for the two included trials (Güven, Sungur, Eser, et al. 2004; Pichamuthu et al. 2010) comparing FFP to conventional therapy is provided in Figure 2(B). In a fixed-effect meta-analysis, there was no significant difference in the incidence of IMS between two compare groups [RR: 1.34, 95% CI (0.655, 2.742), $I^2 = 52.7\%$, $p = 0.146$] (Figure 2(B)).

Table 2. Outcomes in included studies.

Author	Mortality		IMS		Hospital stay (day)		ICU length (day)		Serum cholinesterase level (IU)		Needed ventilation		Ventilation duration (day)	
	N/total (%)		N/total (%)		Mean $\pm$ SD		Mean $\pm$ SD		Mean $\pm$ SD		N/total		N/total	
	Without FFP	With FFP	Without FFP	With FFP	Without FFP	With FFP	Without FFP	With FFP	Without FFP	With FFP	Without FFP	With FFP	Without FFP	With FFP
Güven, Sungur, et al. (2004)	3/21	2/12	6/15	2/10	–	–	8.71 $\pm$ 4.5	9 $\pm$ 3.21	500.2 $\pm$ 262	364.5 $\pm$ 163	–	–	–	–
Pichamuthu et al. (2010)	2/17	4/15	5/14	10/9	9.8 $\pm$ 1.6	12.4 $\pm$ 2	–	–	275 $\pm$ 90	296 $\pm$ 47	14/5	14/5	7.5 $\pm$ 1.5	10 $\pm$ 2.1
Pazooki et al. (2011)	2/17	4/15	–	–	5 $\pm$ 5	28 $\pm$ 3	–	–	–	–	–	–	–	–
Dayananda et al. (2016)	6/34	0/40	–	–	–	–	12.45 $\pm$ 4.13	8.35 $\pm$ 4.3	–	–	36/4	28/12	6.65 $\pm$ 3.52	3.95 $\pm$ 3.05
Saad et al. (2017)	3/32	0/35	–	–	3.06 $\pm$ 1.4	35 $\pm$ 2.43	–	–	2.24 $\pm$ 0.5	2.48 $\pm$ 0.7	–	–	–	–

N: number; IMS: intermediate syndrome; ICU: intensive care unit; FFP: fresh frozen plasma.

Table 3. Characteristics of included studies.

Study	Design	Intoxicated patients (N)	Type of OP	Intervention	Outcomes assessing	Groups		
						FFP group (N)		
Pichamuthu et al. (2010)	Open-labeled, 3-arm randomized trial	60	NM	FFP, albumin, and saline	Incidence of IMS, need for mechanical ventilation, atropine requirement, plasma pseudocholinesterase level, mortality, hospital stay, and ventilation duration	19	19 Saline	20 Albumin
Dayananda et al. (2016)	Randomized clinical trial	80	NM	FFP, atropine, and oximes	Mortality, ICU length, need for mechanical ventilation and ventilation duration, serum cholinesterase level	40	40	40
Güven, Sungur, et al. (2004)	Partially Randomized clinical trial	33	NM	FFP, atropine, and pralidoxime	BuChE levels, incidence of IMS, mortality rates, ICU length	12	21	21
Pazooki et al. (2011)	Randomized clinical trial	56	NM	FFP, atropine, and pralidoxime	Hospitalization length, mortality rates	28	28	28
Saad et al. (2017)	Randomized clinical trial	70	NM	FFP, atropine, and oximes	Cholinesterase level, hospitalization length, mortality rates, ventilation duration	35	35	35

N: number; IMS: intermediate syndrome; ICU: intensive care unit; FFP: fresh frozen plasma; OP: organophosphate poisoning; NM: not mentioned.

Effect of FFP therapy in comparison to conventional regimen on hospitalization stay in patients with OP poisoning

A random meta-analysis results of three trials (Pichamuthu et al. 2010; Pazooki et al. 2011; Saad et al. 2017) indicated, there was no significant difference between FFP-treated groups when compared with conventional therapy in terms of mean hospital stay duration [SMD: -0.106 , 95% CI (-0.434 , 0.223), $I^2 = 90.0$, $p < 0.001$] (Figure 2(C)). We observed inconsistent results and publication bias (Figure 4) in favor of hospital stay duration between FFP therapies compared to the conventional regimen. Based on Begg's ($p = 0.117$) funnel plot and Egger test ($p = 0.112$), we did not detect any meaningful statistical difference.

Effect of FFP therapy in comparison to conventional regimen on ICU length in patients with OP poisoning

A fixed meta-analysis of the results of two trials (Güven, Sungur, Eser, et al. 2004; Dayananda et al. 2016) exhibited that, there was a significant mean difference (MD) between the FFP-treated group compared to the conventional treatment group [SMD: -2.672 , 95% CI (-4.189 , -1.154), $I^2 = 85.8$, $p = 0.008$] (Figure 2(D)). In a random effects meta-analysis, there was no significant MD between the FFP-treated group when compared with conventional therapy [SMD: -2.015 , 95% CI (-6.308 , 2.277), $I^2 = 85.8$, $p = 0.008$] (Figure 2(E)).

Effect of FFP therapy in comparison to conventional regimen on choline esterase level in patients with OP poisoning

A random meta-analysis of the results of three trials (Güven, Sungur, Eser, et al. 2004; Pichamuthu et al. 2010; Saad et al. 2017) showed that at the end of the trials, there was a small MD between FFP-treated group when compared with conventional therapy [SMD: -0.573 , 95% CI (-1.924 , 0.777), $I^2 = 92.2$, $p < 0.001$] (Figure 2(F)) which was also not significant. We observed inconsistent results and publication bias

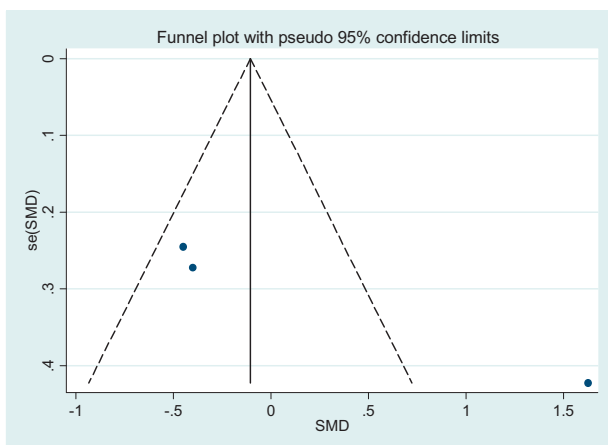


Figure 4. Funnel plot for random effects meta-analysis of mean difference in hospital stay comparing FFP therapy with conventional regimen.

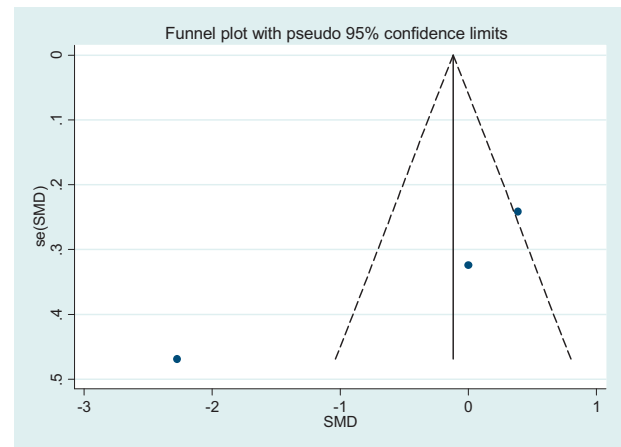


Figure 5. Funnel plot for random effects meta-analysis of mean difference in choline esterase level, comparing fresh frozen plasma therapy with conventional regimen.

(Figure 5) in favor of the choline esterase level between FFP therapy in comparison to conventional therapy, although based on Begg's ($p = 0.117$) funnel plot and Egger test ($p = 0.186$), we did not detect any significant result.

Effect of FFP therapy in comparison to conventional regimen on need for ventilation in patients with OP poisoning

The summary for RR of mechanical ventilation in OP poisoning patients for two selected trials (Pichamuthu et al. 2010; Dayananda et al. 2016) comparing FFP with conventional treatment was assessed. In a fixed-effect meta-analysis, we observed that at the end of the trials, need for ventilation decreased in the FFP-treated group compared with the conventional therapy but this difference was not significant [RR: 0.84 , 95% CI (0.691 , 1.022), $I^2 = 19.8$, $p = 0.264$] (Figure 2(G)).

Effect of FFP therapy in comparison to conventional regimen on ventilation duration in patients with OP poisoning

A random meta-analysis of the results of two trials (Güven, Sungur, Eser, et al. 2004; Saad et al. 2017) showed that at the end of the trials, there was a slight significant MD between FFP-treated group compared with the conventional therapy [SMD: 0.247 , 95% CI (-1.863 , 2.356) $I^2 = 96.0\%$, $p < 0.0001$] (Figure 2(H)).

Discussion

In this study, we found that the addition of FFP to conventional therapy did not improve the outcomes of mortality, IMS, hospital length of stay, cholinesterase levels, need or duration of mechanical ventilation. However, a significant difference on ICU stay was observed in FFP group when compared with conventional therapy.

Advances in studies indicated that a treatment strategy based on FFP therapy may be useful in the treatment of OP poisoning (Doctor and Saxena 2005). The theory of using FFP

against OPs poisoning was initially performed in Güven, Sungur, Eser, et al. (2004), but the results indicated non-remarkable differences in IMS and mortality rates. Pichamuthu et al. (2010) observed that OPs are engaged with BuChE which causes a reduction in the plasma cholinesterase level, though this was not clinically significant. There are different classes of OPs with the same mechanism of action but vary in toxicity. It is suggested that FFP may have a higher effect on WHO Class IA OPs (high toxic OP) such as phorate compared to WHO Class II OPs like dimethoate (less toxic OPs) (World Health Organization 2010). BuChE is highly inhibited by fenthion/chlorpyrifos in comparison with dimethoate (Eddleston et al. 2009). The type of OP may be the factor that the FFP did not have efficacy, as it had not been mentioned in the evaluated studies.

Adding FFP to treatment regimen is not significantly effective on reducing mortality rate in comparison to conventional treatment, which is against the results of the two mentioned studies. Saad et al. (2017) reported the incidence of mortality rate in control and treated groups, as 3.32 (9.3%) and 0.35 (0%), respectively. Conversely, in another study that has been presented by Dayananda et al., such result was noticed and the mortality rate in control group was 6.34 (18%), whereas in FFP group was 0.40 (0%). Another reason for this discrepancy may be explained for different dosage of FFP administration in these studies.

Pichamuthu et al. (2010) presented an IMS incidence of 53% after FFP therapy in the FFP-treated group vs. 26% in the control group. In contrast, a study conducted by Güven, Sungur, Eser, et al. (2004) showed that the IMS incidence was 0% in the FFP-treated group and 25% in the control group, but analysis of the data indicated no statistical difference. The reason for lower IMS rate in the Güven's study may be using repeated doses of FFP for OP poisoning cases.

Successful plasmapheresis of OP-poisoned patients was reported in a few case reports (Güven, Sungur, et al. 2004; Ahila Ayyavoo et al. 2011).

The randomized clinical trial (RCT) carried out by Saad et al. (2017) reported that the addition of FFP therapy improved clinical recovery by declining in mortality rates, hospitalization length, and the need of ICU admission. In addition, Dayananda et al. (2016) stated that FFP therapy decreased the length of ICU stay and ventilation duration. Despite this, our meta-analysis showed no benefit with the addition of FFP to conventional therapy for these outcomes. Whereas in a fixed model analysis, a significant difference on ICU stay was observed in FFP group when compared with conventional therapy but based on random analysis and a significant heterogeneity in this variable, this difference is not important.

A meta-analysis of the published research on efficacy on FFP treatment in OP poisoning in 2014 (Mirfazaelian et al. 2014), only had been included three articles and did not report any meaningful difference in mortality, hospital stay, and IMS syndrome. In our study, the literature search is more extensive than Nafea et al.'s (2019) systematic review. In their study, only PubMed and Scopus (from inception till January 2017) were selected to identify relevant RCTs but PubMed, ScopeMed, Cochrane, Scopus, and Google Scholar databases

as used in this study were not utilized. We have compared the effect of FFP based on several items including the length of hospitalization, length of ICU admission, need for mechanical ventilation and its duration, clinical recovery point, choline esterase level, mortality rate, and IMS occurrence while the other review has only evaluated other outcomes including total atropine dose, total oximes dose, hospital stay length, and death.

As there are only few clinical studies that have compared the effects of FFP therapy with conventional (atropine, oximes/pralidoxime) regimen on OP poisoning patients, future investigations should include larger studies with a standardized reporting regimen and examine the various subgroups of OP compounds to clarify if FFP has a role in the management of OP-poisoned patients. Also in mentioned clinical trials, "FFP time administration" had not been reported. So the delay to FFP injection may be a responsible contributor to this response, as AChE activity takes 14 days to recover in the body (Kobayashi et al. 2007). In addition, the normal AChE activity range is between 4000–11,000 KU/L (Bao et al. 2017), and the level of enzymes may be influenced by factors including age, sex, BMI, and smoking (Karasova et al. 2017; Rathish et al. 2018). Therefore, variability in the activity of AChE among patients before treatment may cause different responses to FFP administration, which should be evaluated in future studies. Finally, the sample sizes in the studies necessitate larger RCTs to evaluate the effect of FFP. However, as it stands the current evidence does not support this hypothesis.

Conclusion

From this systematic review and meta-analysis, the addition of FFP to conventional therapy did not improve the outcomes of mortality, IMS, hospital length of stay, cholinesterase levels, need or duration of mechanical ventilation, and ICU stay.

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Author contributions

FGH and RM generated the idea and design of study. MAS and RN searched the literatures in databases, wrote some part of manuscript. AF and MRM participated in statistical analyses and edited result part. NEM, RM, FGH, GHD, and AW reviewed the manuscript. NEM and RM wrote the discussion section, and with AW and FGH, supervised all parts of revision part of manuscript. RM acted as the corresponding author.

Declaration of interest

The authors take complete accountability for the content of this paper. Affiliations for the authors have been inserted in the cover page. All authors have contributed to the preparation of this article. This paper has had a little financial support to be designed and provided. The authors declare no conflict of interest in this study.

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